

SKAL Bench

Routine & Special Core Analysis

Methods, Equations and References

Technical documentation for all analysis modules

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Generated from the application's own module definitions, so the equations and citations here are those actually evaluated by the software.

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1 Introduction

Every calculation in SKAL Bench is listed here with the equation it evaluates, the measurements it requires and the published source it follows. Nothing is a black box: where a correlation carries fitted coefficients, the defaults are stated and the means of calibrating them to core data is described.

How a test runs

Select a test from the home page. Each lists the measurements it needs and offers a pre-formatted spreadsheet template. Enter one plug by hand or upload a batch, and the console returns the headline result, the supporting figure and the full computed table. Figures export at publication resolution; both the plotted points and the complete table export as CSV. Sample-level fields such as plug length and fluid viscosity are entered once per plug, while row fields are the measurements that vary within a run.

2 Permeability

Steady-State Gas Permeameter

Multi-pressure gas flow, Klinkenberg-corrected

$$k_a = \frac{2000 \mu L P_2 Q}{A (P_1^2 - P_2^2)}$$

$$k_a = k_\infty \left(1 + \frac{b}{P_m} \right), \quad k_\infty = \lim_{1/P_m \rightarrow 0} k_a$$

where:

- k_a = apparent gas permeability at the test mean pressure mD
- k_∞ = Klinkenberg-corrected (equivalent liquid) permeability mD
- Q = volumetric gas flow rate at outlet conditions cm³/s
- μ = gas viscosity cp
- L = plug length cm
- A = plug cross-sectional area cm²
- P_1 = absolute inlet pressure atm
- P_2 = absolute outlet pressure atm
- P_m = mean pressure, $(P_1 + P_2)/2$ atm
- b = gas-slip factor of the Klinkenberg relation atm

Gas is compressible, so the Darcy form uses the squared-pressure difference. Because gas molecules slip at the pore wall, apparent permeability rises as mean pressure falls; measuring at several mean pressures and extrapolating k_a against $1/P_m$ to infinite pressure removes the effect. The intercept, k_∞ , is the Klinkenberg-corrected (equivalent liquid) permeability, and the slope carries the gas-slip factor b .

References. Darcy, H. (1856). "Les Fontaines Publiques de la Ville de Dijon." Dalmont, Paris. Gas slippage: Klinkenberg, L.J. (1941), "The Permeability of Porous Media to Liquids and Gases," API Drilling and Production Practice, 200-213. Procedure: API RP 40 (1998), §6.

Single-Point Gas Permeability

One pressure/flow reading per plug

$$k_a = \frac{2000 \mu L P_2 Q}{A (P_1^2 - P_2^2)}$$

where:

- k_a = apparent gas permeability mD
- Q = volumetric flow rate at outlet conditions cm³/s
- μ = gas viscosity cp
- L = plug length cm
- A = cross-sectional area cm²
- P_1 = absolute inlet pressure atm
- P_2 = absolute outlet pressure atm

A single-point measurement returns the apparent gas permeability at one mean pressure. It carries an uncorrected slip contribution and therefore reads high — always higher than the Klinkenberg value. Use it for screening; use the multi-point steady-state module when the corrected permeability matters.

References. Darcy, H. (1856), as above; API RP 40 (1998), §6.

Pulse-Decay (Unsteady-State)

Low-permeability plugs via pressure decay

$$\alpha = -\frac{d(\ln \Delta P)}{dt}$$

$$k = \frac{1000 \alpha \mu c_f L}{A \left(\frac{1}{V_1} + \frac{1}{V_2} \right)}$$

where:

- α = decay constant, the slope of $\ln \Delta P$ against time s^{-1}
- ΔP = differential pressure across the plug atm
- t = elapsed time s
- k = permeability mD
- μ = fluid viscosity cp
- c_f = fluid compressibility atm^{-1}
- L = plug length cm
- A = cross-sectional area cm^2
- V_1, V_2 = upstream and downstream reservoir volumes cm^3

For tight rock, steady flow is impractically slow. Instead a small pressure pulse is applied across the plug and its decay between two reservoirs is tracked. The decay is exponential, so permeability follows from the slope of $\ln \Delta P$ against time together with the reservoir volumes and fluid compressibility.

References. Brace, W.F., Walsh, J.B. & Frangos, W.T. (1968). "Permeability of Granite under High Pressure." *Journal of Geophysical Research* 73(6), 2225–2236. Practical implementations: Dicker, A.I. & Smits, R.M. (1988), SPE 17578; Jones, S.C. (1997), "A Technique for Faster Pulse-Decay Permeability Measurements in Tight Rocks," *SPE Formation Evaluation* 12(1), 19–26.

Baseline Liquid (Water) Permeability

Coreflood, Darcy steady-state liquid flow

$$k = \frac{1000 Q \mu L}{A \Delta P}$$

where:

- k = liquid permeability mD
- Q = volumetric flow rate cm^3/s
- μ = liquid viscosity cp

- L = plug length cm
- A = cross-sectional area cm²
- ΔP = differential pressure atm

Liquid permeability needs no slip correction, so it is the natural reference for the Klinkenberg-corrected gas value. Fitting several flow rates and forcing the line through the origin is more robust than any single point, and a curved Q- ΔP trend is a warning of turbulence, fines migration or clay swelling rather than a valid Darcy measurement.

References. Darcy, H. (1856), as above; API RP 40 (1998), §6. Forced-origin regression is used because zero differential pressure must give zero flow.

Stress Dependence of Permeability

Correct ambient plug permeability to reservoir net confining stress

$$k(\sigma) = k_0 e^{-c\sigma} \quad \implies \quad \ln k = \ln k_0 - c\sigma$$

where:

- $k(\sigma)$ = permeability at net confining stress σ mD
- k_0 = permeability extrapolated to zero net stress mD
- c = stress-sensitivity coefficient psi⁻¹
- σ = net confining stress (confining minus pore pressure) psi

Permeability measured on an unconfined plug overstates the reservoir value, sometimes by a large factor in tight or microfractured rock. Measure permeability at three or more net confining stresses, and this module fits an exponential decline, reports the stress-sensitivity coefficient, and predicts permeability at your reservoir net stress. Net confining stress is the confining pressure minus the pore pressure — enter that, not the raw confining pressure.

References. American Petroleum Institute (1998). "Recommended Practices for Core Analysis" (API RP 40), §5 — measurement at net confining stress. Stress sensitivity of permeability: Fatt, I. & Davis, D.H. (1952), Trans. AIME 195, 329-336; Jones, F.O. & Owens, W.W. (1980), "A Laboratory Study of Low-Permeability Gas Sands," JPT 32(9), 1631-1640.

Measurement Uncertainty (Darcy k)

Propagate instrument tolerances into an uncertainty band on permeability

$$k = \frac{1000 Q \mu L}{A \Delta P}, \quad A = \frac{\pi D^2}{4}$$

$$\left(\frac{u_k}{k}\right)^2 = \left(\frac{u_Q}{Q}\right)^2 + \left(\frac{u_\mu}{\mu}\right)^2 + \left(\frac{u_L}{L}\right)^2 + \left(\frac{2u_D}{D}\right)^2 + \left(\frac{u_{\Delta P}}{\Delta P}\right)^2$$

$$U_{95} = 2k \left(\frac{u_k}{k}\right)$$

where:

- k = permeability mD
- Q = flow rate cm³/s
- μ = viscosity cp
- L = plug length cm
- D = plug diameter cm
- A = cross-sectional area, $\frac{\pi D^2}{4}$ cm²
- ΔP = differential pressure atm
- u_x = standard uncertainty of quantity x same unit as x
- U_{95} = expanded uncertainty at coverage factor $k=2$ mD

A permeability is only as good as the instruments behind it. Enter each measured quantity together with its uncertainty — the gauge tolerance, flow-meter accuracy, caliper resolution — and this module propagates them through Darcy’s law. Because the model is a pure product, the relative variances simply add, with diameter counting twice since area goes as D^2 . The output is the permeability with its expanded uncertainty at roughly 95% confidence, plus a ranked contribution list showing which instrument is actually limiting your accuracy.

References. JCGM 100:2008, “Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement” (GUM), §5 — first-order propagation of uncertainty for a product-form model, with the expanded uncertainty $U = k \cdot u_c$ for a coverage factor $k = 2$ ($\approx 95\%$ confidence).

3 Porosity & Volumetrics

Porosity & Grain Density

Bulk, pore & grain volume, porosity and grain density by the gravimetric (Archimedes) method

$$V_b = \frac{W_{sat} - W_{imm}}{\rho_f}, \quad V_p = \frac{W_{sat} - W_{dry}}{\rho_f}, \quad V_g = V_b - V_p$$

$$\phi = \frac{V_p}{V_b}, \quad \rho_g = \frac{W_{dry}}{V_g}$$

where:

- V_b = bulk volume cm³
- V_p = pore volume cm³
- V_g = grain volume cm³
- W_{dry} = dry mass of the cleaned, oven-dried plug g
- W_{sat} = saturated mass weighed in air g
- W_{imm} = saturated mass weighed immersed in the fluid g
- ρ_f = density of the saturating fluid g/cm³
- ϕ = porosity fraction
- ρ_g = grain density g/cm³

Enter one row per plug: the dry mass, the mass fully saturated in air, and the saturated mass suspended (immersed) in the saturating fluid. Bulk volume comes from Archimedes buoyancy, pore volume from the mass of imbibed fluid, and grain volume from their difference — giving porosity and grain density per plug plus a dataset mean. Use the same fluid for saturation and immersion and enter its density.

References. American Petroleum Institute (1998). "Recommended Practices for Core Analysis" (API RP 40), 2nd ed. — bulk volume by Archimedes buoyancy and pore volume by the gravimetric (saturation) method.

Helium Porosimetry (Boyle's Law)

Grain volume and porosity by double-cell helium expansion — the routine-lab standard

$$P_1 V_r = P_2 (V_r + V_c - V_g)$$

$$V_g = V_c - V_r \left(\frac{P_1}{P_2} - 1 \right), \quad V_p = V_b - V_g, \quad \phi = \frac{V_p}{V_b}$$

where:

- P_1 = absolute charge pressure in the reference cell psia
- P_2 = absolute pressure after expansion psia
- V_r = reference cell volume cm³
- V_c = empty sample chamber volume cm³
- V_g = grain volume cm³

- V_b = bulk volume cm³
- V_p = pore volume cm³
- ϕ = porosity fraction

The routine-lab method for grain volume. Helium is charged into a calibrated reference cell at pressure P1, then expanded into the sample chamber holding the plug, settling at pressure P2. Boyle's law gives the grain volume directly, and with the plug's bulk volume you get pore volume, porosity and grain density. Enter absolute pressures — if your gauge reads gauge pressure, add atmospheric (≈ 14.7 psi) first.

References. American Petroleum Institute (1998). "Recommended Practices for Core Analysis" (API RP 40), 2nd ed., §4 — Boyle's-law double-cell helium expansion. Helium is used because its small molecule accesses fine pore throats and it does not adsorb appreciably on most reservoir minerals.

4 Capillary Pressure & Wettability

Contact Angle

Sessile drop or captive bubble — measure the angle by tangent lines

$$\gamma_{sv} = \gamma_{sl} + \gamma_{lv} \cos \theta \quad (\text{Young, 1805})$$

$$\theta_{water} = \begin{cases} \theta_{drop} & \text{water/brine drop} \\ 180^\circ - \theta_{drop} & \text{oil drop or gas bubble} \end{cases}$$

where:

- γ_{sv} = solid-vapour interfacial tension mN/m
- γ_{sl} = solid-liquid interfacial tension mN/m
- γ_{lv} = liquid-vapour interfacial tension mN/m
- θ_{drop} = angle measured through the drop or bubble phase degrees
- θ_{water} = contact angle measured through the water phase, used for the wettability bands degrees

Upload a drop or bubble photograph, mark the solid baseline and two points on each side of the drop edge, and the console fits a tangent through each pair and measures its angle to the baseline — the classical goniometric construction, not automated drop-shape fitting, so accuracy depends on careful point placement. Because the tangent is measured through the drop phase, the fluid setup matters: for an oil drop or a captive gas bubble the reported water contact angle is the supplementary angle. The result is given both on the seven-band petroleum wettability scale and as the surface-science hydrophilic/hydrophobic call at 90°.

References. Young, T. (1805). "An Essay on the Cohesion of Fluids." Philosophical Transactions of the Royal Society of London 95, 65–87. Measurement convention: Craig, F.F. (1971), "The Reservoir Engineering Aspects of Waterflooding," SPE Monograph 3. Interpretation: Anderson, W.G. (1986), "Wettability Literature Survey — Part 2: Wettability Measurement," JPT 38(11), 1246–1262. Classification bands: Chilingar, G.V. & Yen, T.F. (1983), Energy Sources 7(1), 67–75, with Morrow, N.R. (1990), JPT 42(12), 1476–1484.

Capillary Pressure (MICP)

Navigation page — no calculation of its own.

Penetrometer selection, then high-pressure intrusion analysis

A navigation page grouping the mercury-injection workflow: the high-pressure intrusion analysis and the penetrometer selection tool. No calculation of its own.

High-Pressure Intrusion Analysis

Upload the AutoPore export, get the full Pc-curve report

$$d = -\frac{4\sigma \cos \theta}{P} \quad (\text{Washburn, 1921})$$

$$k_{Purcell} = \frac{\phi}{8} \sum_i \Delta S_i r_i^2, \quad J(S_w) = \frac{P_c}{\sigma \cos \theta} \sqrt{\frac{k}{\phi}}$$

$$S_b = S_{b\infty} \exp\left(\frac{-G}{\log_{10}(P_c/P_d)}\right), \quad k_{Thomeer} = 3.8068 G^{-1.3334} \left(\frac{S_{b\infty}}{P_d}\right)^{1.8834}$$

where:

- d = pore-throat diameter μm
- r = pore-throat radius μm
- σ = mercury surface tension, 485 by convention dyn/cm
- θ = mercury contact angle, 140° by convention degrees
- P = mercury injection pressure psia
- ϕ = porosity fraction
- ΔS_i = saturation increment of interval i fraction
- $J(S_w)$ = Leverett J-function dimensionless
- S_w = wetting-phase saturation fraction
- P_c = capillary pressure psi
- S_b = mercury saturation as a fraction of bulk volume % BV
- $S_{b\infty}$ = bulk-volume mercury saturation at infinite pressure % BV
- G = Thomeer pore geometrical factor; lower means better-sorted throats dimensionless
- P_d = extrapolated mercury displacement (entry) pressure psia

The full mercury-injection workflow, presented across tabbed sections: instrument summary, intrusion curves, petrophysics, the data table, relative permeability, a custom plot, and the equations behind them. The custom plot tab allows any two of the processed quantities — saturation, pressure, throat radius, pore-size distribution, height above free water, the J-function — to be plotted against one another in a choice of units and on linear or logarithmic axes, with quick presets for the standard views and the same Modify and export controls as the other figures. Washburn's equation converts each injection pressure to a pore-throat diameter, giving the pore-throat size distribution, and the saturation curve yields threshold pressure, r_{35} , sorting, the Leverett J-function, mercury entrapment and a family of permeability estimates (Purcell, Swanson, Winland, Thomeer). The Brooks-Corey fit is taken across the main drainage interval only, excluding the low-pressure conformance region and the high-pressure microporosity tail, neither of which obeys the power law. The Thomeer hyperbola is fitted by searching the displacement pressure with a closed-form linear solve inside, returning the pore geometrical factor G , entry pressure P_d and the extrapolated $S_{b\infty}$.

References. Washburn, E.W. (1921). "Note on a Method of Determining the Distribution of Pore Sizes in a Porous Material." PNAS 7(4), 115–116. Purcell, W.R. (1949), Trans. AIME 186, 39–48. Burdine, N.T. (1953), Trans. AIME 198, 71–78. Leverett, M.C. (1941), Trans. AIME 142, 152–169. Swanson, B.F. (1981), JPT 33(12), 2498–2504. Thomeer, H.J. (1960), JPT 12(3), 73–77, and Thomeer, H.J. (1983), JPT 35(4), 809–814. Brooks, R.H. & Corey, A.T. (1964), Colorado State University Hydrology Paper 3. Winland r_{35} as published in Kolodzie, S. (1980), SPE 9382. Flow-zone indicator: Amaefule, J.O. et al. (1993), SPE 26436.

Penetrometer Selection Tool

Pick the right bulb size and stem volume for your sample

$$\text{stem utilisation} = \frac{V_{\text{intruded}}}{V_{\text{stem}}} \times 100\% \quad (\text{target 25-90\%})$$

where:

- V_{intruded} = mercury volume the plug is expected to take up cm³
- V_{stem} = penetrometer stem capacity cm³

Matches a plug to a penetrometer by estimating the mercury volume it will take up and checking the resulting stem utilisation. The part numbers and bulb/stem volumes are published manufacturer specifications, not measured data.

References. Micromeritics AutoPore penetrometer specifications (published catalogue data, Table 5-1). Selection guidance follows API RP 40 (1998) and standard mercury-porosimetry practice: too little stem utilisation gives poor resolution, too much risks exceeding stem capacity mid-run.

Capillary Pressure (Centrifuge)

Drainage Pc curve from multi-speed centrifuge data (Hassler-Brunner)

$$\omega = \frac{2\pi N}{60}, \quad P_{c1} = \frac{1}{2} \Delta\rho \omega^2 (r_2^2 - r_1^2)$$

$$\bar{S}(P_{c1}) = \int_0^1 S(uP_{c1}) g(u) \, du, \quad g(u) = \frac{\frac{1}{2}(r_1 + r_2)}{\sqrt{r_2^2 - u(r_2^2 - r_1^2)}}$$

$$\text{Hassler-Brunner: } S_1 = \bar{S} + P_{c1} \frac{d\bar{S}}{dP_{c1}}$$

where:

- ω = angular velocity rad/s
- N = rotational speed rev/min
- P_{c1} = capillary pressure at the inner (inlet) face psi
- $\Delta\rho$ = density difference between the two phases g/cm³
- r_1 = radius from the axis to the inner core face cm
- r_2 = radius from the axis to the outer core face cm
- $\bar{S}$ = measured average saturation across the plug fraction
- S_1 = saturation at the inlet face fraction
- u = normalised position variable, P_c/P_{c1} dimensionless
- $g(u)$ = geometry weighting of the centrifugal field dimensionless

Enter one row per centrifuge speed: the rotational speed and the measured equilibrium average water saturation at that speed. The capillary pressure at the inner (inlet) face is computed from the speed and the plug's radial position, and the average saturation is converted to the inlet-face saturation by the Hassler-Brunner first approximation. The result is the drainage capillary-pressure curve, from which the maximum Pc and irreducible saturation are read.

References. Hassler, G.L. & Brunner, E. (1945). "Measurement of Capillary Pressures in Small Core Samples." Trans. AIME 160, 114-123. Inlet-face correction after Forbes, P. (1994), The Log Analyst 35(4), 31-53; see also API RP 40 (1998).

Wettability (Amott / USBM)

Amott-Harvey relative displacement index, plus the USBM wettability index

$$\delta_w = \frac{V_{wsp}}{V_{wt}}, \quad \delta_o = \frac{V_{osp}}{V_{ot}}, \quad I_{AH} = \delta_w - \delta_o$$

$$W_{USBM} = \log_{10} \left(\frac{A_1}{A_2} \right)$$

where:

- δ_w = Amott displacement-by-water ratio dimensionless
- δ_o = Amott displacement-by-oil ratio dimensionless
- V_{wsp} = oil displaced by spontaneous water imbibition cm³
- V_{wt} = total oil displaced by water, spontaneous plus forced cm³
- V_{osp} = water displaced by spontaneous oil imbibition cm³
- V_{ot} = total water displaced by oil, spontaneous plus forced cm³
- I_{AH} = Amott-Harvey index, \$+1\$ strongly water-wet to \$-1\$ strongly oil-wet dimensionless
- A_1 = area under the oil-drive capillary pressure curve dimensionless
- A_2 = area under the brine-drive curve dimensionless
- W_{USBM} = USBM wettability index dimensionless

Enter, per sample, the oil volume displaced by spontaneous water imbibition and by total water displacement, and the water volume displaced by spontaneous oil imbibition and by total oil displacement. The console returns the Amott displacement-by-water and displacement-by-oil ratios and the Amott-Harvey index (+1 strongly water-wet to −1 strongly oil-wet). Enter the areas under the two forced-displacement Pc curves as well and the USBM index is reported alongside.

References. Amott, E. (1959). "Observations Relating to the Wettability of Porous Rock." Trans. AIME 216, 156-162. USBM: Donaldson, E.C., Thomas, R.D. & Lorenz, P.B. (1969). "Wettability Determination and Its Effect on Recovery Efficiency." SPE Journal 9(1), 13-20. Review & bands: Anderson, W.G. (1986). "Wettability Literature Survey — Part 2." JPT 38(11), 1246-1262.

5 Relative Permeability

Relative Permeability (Steady-State)

Two-phase oil-water kro/krw curves

$$k_{ro} = \frac{1000 Q_o \mu_o L}{A \Delta P k_{abs}}, \quad k_{rw} = \frac{1000 Q_w \mu_w L}{A \Delta P k_{abs}}$$

where:

- k_{ro} = relative permeability to oil fraction
- k_{rw} = relative permeability to water fraction
- Q_o, Q_w = oil and water flow rates cm³/s
- μ_o, μ_w = oil and water viscosities cp
- L = plug length cm
- A = cross-sectional area cm²
- ΔP = differential pressure atm
- k_{abs} = absolute permeability of the plug mD

Both phases are injected together at fixed ratios until pressure and production stabilise, and Darcy's law is applied to each phase in turn. Steady state is slower than an unsteady-state flood but far less sensitive to numerical differentiation, so it is the reference method where accuracy matters.

References. Hassler, G.L. (1944), US Patent 2,345,935. Method and interpretation: Osoba, J.S., Richardson, J.G., Kerver, J.K., Hafford, J.A. & Blair, P.M. (1951), "Laboratory Measurements of Relative Permeability," Trans. AIME 192, 47-56; Richardson, J.G. et al. (1952), Trans. AIME 195, 187-196.

Relative Permeability (Unsteady-State — JBN)

kro/krw from a single waterflood displacement

$$k_{ro} = f_o \frac{d(1/Q_{iD})}{d\left(\frac{1}{Q_{iD} I_R}\right)}, \quad k_{rw} = (1 - f_o) \frac{d(1/Q_{iD})}{d\left(\frac{1}{Q_{iD} I_R}\right)}$$

where:

- k_{ro} = relative permeability to oil fraction
- k_{rw} = relative permeability to water fraction
- f_o = fractional flow of oil at the outlet face fraction
- Q_{iD} = cumulative water injected, in pore volumes dimensionless
- I_R = relative injectivity, the baseline pressure drop divided by the current one dimensionless

References. Johnson, E.F., Bossler, D.P., and Naumann, V.O. (1959). "Calculation of Relative Permeability from Displacement Experiments." Petroleum Transactions, AIME, Vol. 216, pp. 370-372.

Brooks-Corey Fit

Fit a Corey-type power-law model + fractional flow

$$S_e = \frac{S_w - S_{wc}}{1 - S_{wc} - S_{or}}$$

$$k_{rw} = k_{rw}^{max} S_e^{N_w}, \quad k_{ro} = k_{ro}^{max} (1 - S_e)^{N_o}$$

where:

- S_e = normalised (effective) water saturation fraction
- S_w = water saturation fraction
- S_{wc} = connate (irreducible) water saturation fraction
- S_{or} = residual oil saturation fraction
- k_{rw}^{max} = water relative permeability endpoint fraction
- k_{ro}^{max} = oil relative permeability endpoint fraction
- N_w, N_o = Corey exponents for water and oil dimensionless

References. Corey, A.T. (1954). "The Interrelation Between Gas and Oil Relative Permeabilities." Producers Monthly, 19(1), 38-41. Brooks, R.H. and Corey, A.T. (1964). "Hydraulic Properties of Porous Media." Hydrology Papers No. 3, Colorado State University.

Relative Permeability

Navigation page — no calculation of its own.

Steady-state, unsteady-state (JBN), or Brooks-Corey fit

A navigation page grouping the three relative-permeability routes: steady-state, unsteady-state (JBN) and the Brooks-Corey fit/prediction screen. No calculation of its own.

6 Electrical Properties

Formation Factor & Cementation (a, m)

Fit $F = a \cdot \phi^{-m}$ across brine-saturated plugs to get the formation-factor law

$$F = \frac{R_o}{R_w} = a \phi^{-m} \quad \Rightarrow \quad \log F = \log a - m \log \phi$$

$$R_{w2} = R_{w1} \frac{T_1 + 21.5}{T_2 + 21.5} \quad (\text{Arps, 1953; } T \text{ in } ^\circ\text{C})$$

where:

- F = formation resistivity factor dimensionless
- R_o = resistivity of the plug at 100\{\}% brine saturation $\Omega \cdot \text{m}$
- R_w = brine resistivity at the measurement temperature $\Omega \cdot \text{m}$
- a = tortuosity (lithology) factor dimensionless
- m = cementation exponent dimensionless
- ϕ = porosity fraction
- T_1, T_2 = temperatures for the Arps correction $^\circ\text{C}$

Enter one row per brine-saturated plug: its porosity and its resistivity at 100% brine saturation (R_o). With the brine resistivity (R_w), each plug's formation factor $F = R_o/R_w$ is computed, then a straight line is fitted to $\log F$ versus $\log \phi$. The slope gives the cementation exponent m and the intercept the tortuosity/lithology factor a (Archie's original $a = 1$; Humble formula $a \approx 0.62$, $m \approx 2.15$ for sandstones).

References. Archie, G.E. (1942). "The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics." Trans. AIME 146, 54-62. Tortuosity/lithology factor a : Winsauer, W.O., Shearin, H.M., Masson, P.H. & Williams, M. (1952). "Resistivity of Brine-Saturated Sands in Relation to Pore Geometry." AAPG Bulletin 36(2), 253-277.

Resistivity Index & Saturation Exponent (n)

Fit $I_R = S_w^{-n}$ across a desaturation series to get Archie's saturation exponent

$$I_R = \frac{R_t}{R_o} = S_w^{-n} \quad \Rightarrow \quad \log I_R = -n \log S_w$$

$$\hat{n} = -\frac{\sum_i x_i y_i}{\sum_i x_i^2}, \quad x = \log S_w, \quad y = \log I_R \quad (\text{forced through the origin})$$

where:

- I_R = resistivity index dimensionless
- R_t = resistivity at the stated water saturation $\Omega \cdot \text{m}$
- R_o = resistivity at 100\{\}% brine saturation $\Omega \cdot \text{m}$
- S_w = water saturation fraction
- n = saturation exponent dimensionless

Enter one row per desaturation step on a single plug: the water saturation and the resistivity R_t at that saturation. With the plug's resistivity at 100% brine (R_o), the resistivity index $IR = R_t/R_o$ is computed and a straight line is fitted to $\log IR$ versus $\log S_w$. The negative slope is Archie's saturation exponent n (≈ 2 for clean water-wet rock; higher for oil-wet or fractured systems).

References. Archie, G.E. (1942). "The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics." Trans. AIME 146, 54-62 (the second Archie relation). Saturation-exponent dependence on wettability: Keller, G.V. (1953) and Sweeney, S.A. & Jennings, H.Y. (1960).

Waxman-Smiths (Shaly Sand)

Clay-corrected formation factor F^* and cementation exponent m^* for clay-bearing rock

$$C_o = \frac{1}{F^*} (C_w + B Q_v) \quad \Rightarrow \quad F^* = R_o \left(\frac{1}{R_w} + B Q_v \right)$$

$$F^* = a^* \phi^{-m^*}$$

where:

- C_o = conductivity of the brine-saturated rock S/m
- C_w = brine conductivity, $\$/R_w\$ S/m$
- F^* = intrinsic (clay-corrected) formation factor dimensionless
- B = equivalent counterion conductance S·cm²/meq
- Q_v = cation exchange capacity per unit pore volume meq/cm³
- R_o = resistivity at 100% brine Ω·m
- R_w = brine resistivity Ω·m
- a^*, m^* = clay-corrected tortuosity factor and cementation exponent dimensionless
- ϕ = porosity fraction

In clay-bearing sandstone the clay's counterion conductivity short-circuits the rock, so the apparent Archie formation factor is too low and the cementation exponent comes out wrong. Waxman-Smiths separates the two conduction paths: enter each plug's porosity, R_o , and cation exchange capacity per unit pore volume (Q_v), and the console removes the clay contribution to give the intrinsic F^* , then fits $F^* = a^* \phi^{-m^*}$. The clean-Archie fit is reported alongside so you can see how much the clay was biasing it.

References. Waxman, M.H. & Smits, L.J.M. (1968). "Electrical Conductivities in Oil-Bearing Shaly Sands." SPE Journal 8(2), 107-122. Counterion conductance B correlation: Juhász, I. (1981), SPWLA 22nd Annual Logging Symposium. Compare with the clean-sand case of Archie, G.E. (1942), Trans. AIME 146, 54-62.

7 NMR Petrophysics

NMR Porosity & Permeability

Fit T2 decay (ExpDec3), derive ϕ , CBW, BVI, FFI, and Coates/SDR permeability

$$\phi_{NMR} = \frac{A \cdot K}{V_b}, \quad T_{2LM} = \exp\left(\frac{\sum_i f_i \ln T_{2i}}{\sum_i f_i}\right)$$

$$k_{SDR} = a \phi^m T_{2LM}^n \quad (\text{Kenyon et al., 1988})$$

$$k_{Coates} = \left(\frac{\phi}{C}\right)^p \left(\frac{FFI}{BVI}\right)^q \quad (\text{Timur-Coates})$$

where:

- ϕ_{NMR} = NMR porosity fraction
- A = total signal amplitude of the echo train instrument units
- K = instrument calibration constant cm³ per unit amplitude
- V_b = plug bulk volume cm³
- T_{2LM} = logarithmic mean of the T_2 distribution ms
- f_i = amplitude of distribution bin i fraction
- T_{2i} = relaxation time of bin i ms
- BVI = bulk volume irreducible, signal below the T_2 cutoff fraction
- FFI = free fluid index, signal above the cutoff fraction
- C = Timur-Coates calibration constant dimensionless
- a, m, n = SDR coefficient and exponents dimensionless
- p, q = Timur-Coates exponents dimensionless

A CPMG echo train is inverted two ways — a three-component exponential fit and a regularised inverse Laplace transform — and the amplitude is converted to porosity through the instrument calibration constant and the plug bulk volume. The distribution is split at the clay-bound and BVI cutoffs to give CBW, BVI and FFI, from which the SDR and Timur-Coates permeabilities follow. Two calibration controls are provided because the generic literature coefficients routinely miss measured permeability by a factor of several: the SDR coefficient can be back-solved from a measured core permeability, and the BVI cutoff can be derived from a measured irreducible saturation instead of the 33 ms default. Coates is reported as not applicable when almost no signal falls below the BVI cutoff, since the FFI/BVI ratio is then unstable. The echo train is sampled logarithmically before inversion so that the early echoes carrying the clay-bound signal are preserved.

References. SDR model: Kenyon, W.E., Day, P.I., Straley, C. & Willemsen, J.F. (1988), "A Three-Part Study of NMR Longitudinal Relaxation Properties of Water-Saturated Sandstones," SPE Formation Evaluation 3(3), 622-636. Timur-Coates model: Timur, A. (1968), JPT 20(6), 775-786; Coates, G.R., Xiao, L. & Prammer, M.G. (1999), "NMR Logging: Principles and Applications," Halliburton. T2 cutoffs: Straley, C., Rossini, D., Vinegar, H., Tutunjian, P. & Morriss, C. (1997), The Log Analyst 38(2), 84-94. Clay-bound water: Prammer, M.G. et al. (1996), The Log Analyst 37(6).

8 Correlations & Trends

Porosity-Permeability Correlation

Fit $k = f(\phi)$ across your core dataset

$$k = a \phi^b \quad (\text{power law}), \quad k = a e^{b\phi} \quad (\text{exponential})$$

where:

- k = permeability mD
- ϕ = porosity fraction
- a, b = fitted coefficients of the chosen model dimensionless

Fits both a power-law and an exponential trend to a porosity-permeability dataset and recommends the better model by R^2 . The purpose is to extend a handful of core measurements across uncored intervals; the fit is only as good as the rock-type consistency of the dataset, so a poor R^2 usually means several flow units have been mixed rather than that the regression failed.

References. Standard log-linear regression of core porosity-permeability pairs. Interpretation follows Tiab, D. & Donaldson, E.C., "Petrophysics: Theory and Practice of Measuring Reservoir Rock and Fluid Transport Properties," Gulf Professional Publishing. For pore-geometry-based rock typing, see the Rock Typing module.

Rock Typing (Winland r35 / FZI)

Pore-throat radius and flow-zone indicator from routine ϕ - k data

$$\log_{10} r_{35} = 0.732 + 0.588 \log_{10} k - 0.864 \log_{10} \phi \quad (k \text{ in mD}, \phi \text{ in } \%)$$

$$RQI = 0.0314 \sqrt{\frac{k}{\phi}}, \quad \phi_z = \frac{\phi}{1 - \phi}, \quad FZI = \frac{RQI}{\phi_z}$$

where:

- r_{35} = pore-throat radius at 35% mercury saturation μm
- k = permeability mD
- ϕ = porosity (fraction in RQI , percent in the r_{35} regressions) see note
- RQI = reservoir quality index μm
- ϕ_z = normalised porosity, $\phi/(1-\phi)$ dimensionless
- FZI = flow zone indicator μm

Turns an ordinary porosity-permeability table into rock types. For each plug the console computes the Winland r35 pore-throat radius (the throat size at 35% mercury saturation, which best correlates with flow), the Pittman r35 regression for comparison, and the reservoir quality index and flow-zone indicator. Plugs sharing an FZI belong to the same hydraulic flow unit and should fall on one line in the RQI - ϕ_z plot, which is the basis for permeability prediction in uncured intervals.

References. Winland r_{35} as published in Kolodzie, S. (1980), SPE 9382; refined regressions in Pittman, E.D. (1992), "Relationship of Porosity and Permeability to Various Parameters Derived from Mercury Injection-Capillary Pressure Curves for Sandstone," AAPG Bulletin 76(2), 191-198. Flow-zone indicator: Amaefule, J.O., Altunbay, M., Tiab, D., Kersey, D.G. & Keelan, D.K. (1993), SPE 26436. Pore-size classes after Hartmann, D.J. & Beaumont, E.A. (1999), AAPG Treatise.

9 Methods and Conventions

Calibration philosophy

Every empirical permeability transform in this software — SDR, Timur-Coates, Swanson, Winland, Thomeer — carries coefficients that were fitted to a particular set of rocks. Applied unchanged to a different formation they are commonly wrong by a factor of several, and that is a property of the correlations, not a defect of the measurement. Wherever it is possible the console therefore exposes the coefficient and provides a way to anchor it to your own core data, and reports the generic literature value alongside so the size of the correction is visible.

Curve smoothing

$$y_i^* = \frac{1}{35} (-3y_{i-2} + 12y_{i-1} + 17y_i + 12y_{i+1} - 3y_{i+2}) \quad (5\text{-point quadratic})$$

Every chart offers Savitzky-Golay smoothing at 5, 7 or 9 points. The quadratic Savitzky-Golay kernel is used in preference to a moving average because it preserves peak height and position, which matters where the peak is itself the measurement — T2 distributions and pore-throat spectra. Endpoints are left unsmoothed rather than extrapolated. Smoothing is applied only to what is drawn: every reported value, fitted parameter and R² is computed from the raw data, and a note appears beneath any chart that is being displayed smoothed.

Reference. Savitzky, A. & Golay, M.J.E. (1964). "Smoothing and Differentiation of Data by Simplified Least Squares Procedures." *Analytical Chemistry* 36(8), 1627-1639.

Numerical differentiation

The JBN relative-permeability calculation differentiates the production and pressure data twice, which strongly amplifies scatter — particularly just after breakthrough. A five-point Savitzky-Golay smoother is therefore applied to the monotone series before differentiation, and can be switched off from the input fields. Points that come out negative or otherwise unphysical are filtered rather than plotted.

Brine resistivity and temperature

$$R_{w2} = R_{w1} \frac{T_1 + 21.5}{T_2 + 21.5} \quad (T \text{ in } ^\circ\text{C})$$

Brine resistivity is strongly temperature-dependent, so it must be stated at the same temperature as the resistivity measurements before a formation factor is formed. The electrical modules apply the Arps correction, $R_{w2} = R_{w1} \cdot (T_1 + 21.5) / (T_2 + 21.5)$ with temperatures in °C, and report when a correction has been made. The resistivity index needs no such correction because it is a ratio taken on the same plug, so temperature cancels.

Reference. Arps, J.J. (1953). "The Effect of Temperature on the Density and Electrical Resistivity of Sodium Chloride Solutions." *Journal of Petroleum Technology* 5(10), 17-20.

Constrained regression

$$\hat{n} = -\frac{\sum x_i y_i}{\sum x_i^2} \quad \text{vs.} \quad \text{unconstrained } (\hat{a}, \hat{n})$$

Two of the Archie fits have a physically fixed point and are fitted accordingly. The resistivity index must equal one at full water saturation, so the saturation exponent is estimated with the line forced through the origin in log-log space; the unconstrained fit is reported alongside, because a large intercept is a useful warning that the data are noisy or non-Archie. The formation factor is reported both with a free tortuosity factor and with Archie's original $a = 1$ constraint.

Centrifuge inversion

$$\bar{S}(P_{c1}) = \int_0^1 S(uP_{c1}) g(u) \, du$$

Converting an average saturation measured in a centrifuge into the saturation at the inlet face is an inverse problem. The Hassler-Brunner first approximation is exact only in the limit where the plug is short relative to its distance from the axis. This console also solves the underlying integral equation directly, as a regularised least-squares problem with a monotonicity constraint, and selects between the two on the basis of the $r1/r2$ ratio — round-trip tests on synthetic curves show the direct inversion to be substantially more accurate for long plugs and the Hassler-Brunner approximation better as $r1/r2$ approaches one. Both values are always reported.

Reference. Hassler, G.L. & Brunner, E. (1945), Trans. AIME 160, 114-123; Forbes, P. (1994), The Log Analyst 35(4), 31-53.

Measurement uncertainty

$$\left(\frac{u_k}{k}\right)^2 = \sum_i \left(\frac{\partial \ln k}{\partial \ln x_i}\right)^2 \left(\frac{u_{x_i}}{x_i}\right)^2, \quad U_{95} = 2u_k$$

Darcy's law is a product of measured quantities, so relative variances add and the plug diameter counts twice because area goes as the square of diameter. The uncertainty module propagates instrument tolerances on this basis and reports the expanded uncertainty at a coverage factor of two, together with a ranking of which instrument dominates the error budget.

Reference. JCGM 100:2008. "Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement" (GUM), §5.

Data export

Two exports are offered, and they differ in scope. Plotted points (CSV), beneath each figure, contains only the points drawn on that chart. Full results (CSV), on the result card, contains the complete computed table for every row, including the intermediate columns the chart never shows — for the centrifuge module, for instance, it carries both the Hassler-Brunner and the direct-inversion saturations side by side. When a chart is displayed smoothed the exported filename records the smoothing window,

so a smoothed export cannot be mistaken for raw data. Figures export separately at publication resolution.

Modifying a figure

Every chart carries a Modify control that opens a panel for presentation changes: a title or sample name, axis labels, axis minima and maxima, and gridlines on or off. A live preview shows the effect before anything is exported, and the title is drawn inside the exported image rather than only on screen. Axis limits clip the view alone — the fit and every reported value continue to use the whole dataset, so a figure can be zoomed to a region of interest without altering the result behind it.

User-applied trend lines

On charts of discrete measurements the same panel offers a trend line: the module's own fit, no line at all, or a linear, power or exponential fit applied by the user, with the equation and R^2 shown beneath the plot and carried into the export. R^2 is computed against the untransformed measurements rather than in the space each model linearises to, so the three models can be compared on equal terms. A trend chosen here is an annotation on the figure and never feeds the reported result, which is always computed by the module's own method from the raw data. The control is withheld on charts of continuous curves and distributions — relative permeability, T2 spectra — where a trend line would not be meaningful.

When a calculation fails

If a screen fails it is replaced by an error panel rather than a blank page, and the rest of the application stays usable. The panel shows a reference code derived from the fault itself, so the same problem always produces the same code, together with a report — version, module, error and timestamp — that can be sent in one action. Nothing entered is transmitted anywhere: all calculation happens in the browser, and a report contains only what the user chooses to send.

Units

Each input field states its expected unit and the console converts internally; permeability is reported in millidarcy, pressure in psi or atm as labelled, lengths in centimetres and viscosity in centipoise. Saturations and porosity are fractions, not percentages, except where a field is explicitly labelled as a percentage. A separate unit converter is provided under the Converter tab, and the conversion factors it uses are exact definitions rather than rounded values.

10 Limitations

- Empirical permeability transforms are only as good as their calibration. Treat any uncalibrated permeability from NMR or mercury injection as an order-of-magnitude estimate until it is anchored to measured core.
- Fixed T2 and saturation cutoffs are regional conventions. The 33 ms sandstone and 92–100 ms carbonate values are Gulf of Mexico defaults and should be confirmed against core wherever possible.
- The Timur-Coates model becomes unstable as bound water approaches zero, and is reported as not applicable in clean, near-irreducible rock rather than returning a spurious number.
- The contact-angle tool is a manual tangent construction, so its accuracy depends on point placement; it is not automated drop-shape analysis.
- Fits are ordinary least squares in the stated transform space unless a physical constraint is noted, and do not carry confidence intervals except in the dedicated uncertainty module.
- Ambient plug measurements overstate reservoir permeability. Use the stress-dependence module where net confining stress matters.

11 Standards

Laboratory procedures follow the American Petroleum Institute's *Recommended Practices for Core Analysis* (API RP 40, 2nd edition, 1998), the reference standard for routine and special core analysis. Uncertainty is expressed following JCGM 100:2008 (GUM). Individual module citations are listed with each entry above.

Results are engineering estimates that depend on input data and local calibration. Confirm important values against core measurements before relying on them. Provided as is, without warranty.